

Dini N Kathriarachchi

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EDUCATION

PRINCETON UNIVERSITY, New Jersey, USA (GPA: 3.7/4.0) **Expected Graduation: Jun 2027**
B.S.E. in Mechanical and Aerospace Engineering || Minors in Computer Science & Robotics & Sustainable Energy
Relevant Coursework: Energy Tech.; Engineering Design; Robotics; Heat Transfer; Control Systems; Fluids; Structures; Material Science; Algorithms; Data Science; Microcomputers for Measurement & Control; Electricity Sector

WORK EXPERIENCE

AI Integration Intern, Robotics & AI Team (Divisional Corp.) – Larsen & Toubro **Jun 2026 - Present**

- Conceived 12 high-value computer vision use cases across project sites to improve Safety, Quality & Progress Tracking
- Developing comprehensive strategy and road-map to expand CV adoption by mapping existing physical AI assets, evaluating CV platforms, open-source CV frameworks & edge inference hardware
- Assessing build-vs-buy-vs-partner frameworks and strategizing deployment architecture for 100+ construction sites

Nuclear Research Intern – Princeton Plasma Physics Laboratory **Jun 2025 - Oct 2025**

- Assisted mentor to run nuclear fusion prototype machine PFRC-2, and analyze data from oscilloscopes, gauges & sensors
- Analyzed data using MATLAB & Python, compiled weekly presentations and produced detailed final technical reports
- Synthesized findings into two complete research papers related to characterization of plasma behavior

PROJECTS & ACTIVITIES

Princeton University Robotics Club – Multiple Projects & Roles **Sep 2023 - Present**

- ◇ **Social Chair:** Overseeing the organization, staffing, set up & running of all social events for the club
- ◇ **Hoverloot Project, Technical Lead:** Designing a helium-electric hybrid drone prototype for carrying loads in cost effective and energy efficient manner
- ◇ Working with mechanical (structural frames, motors, landing gear) and electrical components (power distribution boards, electronic speed controllers, batteries) to modify & integrate existing drone designs with airship structures
- ◇ **Wall-E Project, Mechanical Team:** Contributed to preliminary structural & mechanical design of treads and transport mechanisms of robot capable of compact self-storage and automated trash collection
- ◇ **PacBot Project, Hardware Team:** Analyzed previous models & their shortcomings to develop an enhanced small-scale fully autonomous robot able to quickly and effectively navigate a narrow maze
- ◇ Integrated electrical components such as printed circuit boards, connecting wires & sensors into mechanical designs

Autonomous Locomotive Roller Coaster System (Microprocessors Course) **Feb 2026 - May 2026**

- Designed end-to-end system integration and debugging across mechanical, electrical, and software subsystems as part of a 3-person team, diagnosing faults and iterating on the design to deliver a fully functional autonomous demo
- Engineered a 12-support spiral track system for a model train through a 510° helical descent at a 2.4° incline, using Onshape to design various structurally distinct support types with interlocking joints and slanted mounting platforms
- Built a custom embedded controller around a 6502-based motherboard and Arduino processor, including a GAL address-decoding chip, Hall Effect sensors, and a trickle-charge circuit to safely bridge 24V track signals to 5V logic

Fluid & Heat Flow Simulation Project (Heat Transfer Course) **Nov 2025 - Dec 2025**

- Constructed a 2D streamfunction-vorticity finite-difference CFD solver in MATLAB to simulate low-Reynolds-number convective flow and heat transfer around three aircraft wing planform (rectangular, elliptical, tapered)
- Validated simulation accuracy against peer-reviewed literature, achieving Strouhal and Nusselt numbers within small margins of published benchmarks, then applied model to compare vortex-shedding and convective heat-loss behavior

Wing Design Project, Group Lead (2nd-year Mechanical Design Course) **Mar 2025 - May 2025**

- Led team of 8 to successfully design, construct, and test an efficient airfoil capable of withstanding a 2200 lbf-in torque with minimal deflection and a weight of 18.5 oz
- Designed components using CAD (PTC Creo), generated engineering drawings & reports with justified design decisions, detailed specifications & tolerances, and conducted structural analysis simulations (FEA) to optimize design
- Manufactured the components using Computer-Aided Manufacturing (CAM) software and manufacturing techniques such as Computer Numerical Control (CNC), Selective Laser Sintering (SLS) & water jet cutters
- Worked with co-lead to distribute tasks, ensure cooperation among group members, and facilitate communication and coordination with instructional staff to ensure all project expectations and deadlines were met

TECHNICAL SKILLS

Programming: Java, Python (pandas, tensorflow, sklearn), MATLAB, Simulink || **Engineering:** PTC Creo (CAD, FEA, CAM), OnShape, Ansys || **Other:** Ardupilot, LaTeX, QGIS, InDesign, MS Clipchamp || **Language:** German (Elementary)